KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5073A | 5 Credits: 4 | Program Elective | Source-grounded

Petroleum Refining &

 To impart the basic knowledge on fossil fuels, up-stream and downstream processes

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5073A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5073A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Chemical Engineering
Course code	5073A
Course title	Petroleum Refining &
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the basic knowledge on fossil fuels, up-stream and downstream processes
- for crude petroleum.
- To comprehend the manufacturing processes and applications of various
- petrochemical products.

Course outcomes

CO	Official outcome	Study level
CO1	14 Understanding processes of crude petroleum. Illustrate the downstream processes for crude	See official syllabus
CO2	17 Understanding petroleum. Summarize the treatment techniques for petroleum	See official syllabus
CO3	15 Understanding fractions. Summarize the manufacturing processes and	See official syllabus
CO4	14 Understanding applications of various petrochemicals.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Summarise fossil fuels and the up-stream processes of crude petroleum.	4	As specified
Module 2	Illustrate the downstream processes for crude petroleum.	5	As specified
Module 3	Summarize the treatment techniques for petroleum fractions.	4	As specified
Module 4	petrochemicals.	4	As specified

MODULE 1

Summarise fossil fuels and the up-stream processes of crude petroleum.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise fossil fuels and the up-stream processes of crude petroleum.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Classify fossil fuels. 2 Remembering Summarize the production methods,

Classify fossil fuels. 2 Remembering Summarize the production methods, is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify fossil fuels. 2 Remembering Summarize the production methods, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify fossil fuels. 2 remembering summarize the production methods, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify fossil fuels. 2 Remembering Summarize the production methods, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] Classify fossil fuels. 2 Remembering Summarize the production methods, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of

4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding characteristics and properties of gaseous fuels. summarise the characterization and properties of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 1-[illegible] 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding crude petroleum. Illustrate the upstream processes for crude

4 Understanding crude petroleum. Illustrate the upstream processes for crude is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding crude petroleum. Illustrate the upstream processes for crude using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding crude petroleum. illustrate the upstream processes for crude should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding crude petroleum. Illustrate the upstream processes for crude means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-[illegible] 4 Understanding crude petroleum. Illustrate the upstream processes for crude [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India

4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding petroleum. classification of fuel- categorize primary & secondary solid, liquid and gaseous fuels with examples. identify the composition, properties and uses of natural gas, - illustrate the production of natural gas - comprehend the production of shale gas. origin& formation of petroleum - classification of crude oil – properties of crude oil -api gravity and density, viscosity, sulphur content and vapour pressure – evaluation of crude oil. exploration and drilling of crude oil - gas-oil separation – dehydration - desalting- crude stabilization and enhanced oil recovery – sources of indigenous crude - sources of imported crude - petroleum industries in india should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin&

formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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Summarise fossil fuels and the up-stream processes of crude petroleum.
  M1.01     Classify fossil fuels.                                    2
Remembering
            Summarize the production methods,
  M1.02                                                               4
Understanding
            characteristics and properties of gaseous fuels.
            Summarise the characterization and properties of
  M1.03                                                               4
Understanding
            crude petroleum.
            Illustrate the upstream processes for crude
  M1.04                                                               4
Understanding
            petroleum.

Contents:
Classification of fuel- Categorize primary & secondary solid, liquid and gaseous
fuels with
examples.
Identify the composition, properties and uses of Natural gas, - Illustrate the
production of
natural gas - Comprehend the production of shale gas.
Origin& formation of petroleum - Classification of crude oil – Properties of
crude oil -API
gravity and density, viscosity, sulphur content and vapour pressure – Evaluation
of crude
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

support notes.

MODULE 2

Illustrate the downstream processes for crude petroleum.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the downstream processes for crude petroleum.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by

Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate modern refinery operations. 1 understanding illustrate petroleum products from crude oil by should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding fractional distillation. Outline various cracking and reforming processes

4 Understanding fractional distillation. Outline various cracking and reforming processes is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding fractional distillation. Outline various cracking and reforming processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding fractional distillation. outline various cracking and reforming processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding fractional distillation. Outline various cracking and reforming processes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding fractional distillation. Outline various cracking and reforming processes □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating

4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding of petroleum fractions summarise visbreaking, coking, hydro treating should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding processes. Summarise Alkylation, Isomerisation and

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Concept and explanation

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Study points

- Define or identify 4 Understanding processes. Summarise Alkylation, Isomerisation and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding processes. summarise alkylation, isomerisation and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding processes. Summarise Alkylation, Isomerisation and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ 4 Understanding processes. Summarise Alkylation, Isomerisation and □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation – Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes.

4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation – Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes. is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation – Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation

process and Catalytic polymerization processes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding polymerization processes block diagram of a modern refinery operation (including adu, vdu, gas processing, hydrotreater, fccu, isomeriser, alkylation, blending, dcu and sulphur recovery unit) illustrate crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - list the petroleum products from crude oil refining and their boiling ranges. illustrate: cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - catalytic reforming illustrate: visbreaking operation – coking - hydro treating illustrate: sulphuric acid alkylation process, low temperature isomerisation process and catalytic polymerization processes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation – Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation – Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the downstream processes for crude petroleum.
  M2.01     Illustrate modern refinery operations.                  1
Understanding
            Illustrate petroleum products from crude oil by
  M2.02                                                             4
Understanding
            fractional distillation.
            Outline various cracking and reforming processes
  M2.03                                                             4
Understanding
            of petroleum fractions
            Summarise Visbreaking, Coking, hydro treating
  M2.04                                                           4
Understanding
            processes.
            Summarise Alkylation, Isomerisation and
  M2.05                                                           4
Understanding
            polymerization processes
            Series Test – I

Contents:
Block diagram of a modern refinery operation (Including ADU, VDU, gas processing,
hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery
unit)
Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric
distillation
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize the treatment techniques for petroleum fractions.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the treatment techniques for petroleum fractions.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding petroleum fractions. Summarize the purification and treatment

2 Understanding petroleum fractions. Summarize the purification and treatment is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding petroleum fractions. Summarize the purification and treatment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding petroleum fractions. summarize the purification and treatment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding petroleum fractions. Summarize the purification and treatment means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 2 Understanding petroleum fractions. Summarize the purification and treatment [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum

6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding processes of petroleum fractions. summarize the important properties of petroleum should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding products. Outline the experimental methods for determining

2 Understanding products. Outline the experimental methods for determining is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding products. Outline the experimental methods for determining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding products. outline the experimental methods for determining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding products. Outline the experimental methods for determining means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-[illegible] 2 Understanding products. Outline the experimental methods for determining [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various

5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding the properties of petroleum products. physical and chemical impurities in petroleum fractions. purification and treatment process - sweetening process of petroleum fractions - lpg merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. properties of petroleum fractions (api gravity - watson characterization factor - viscosity - sulfur content - true boiling point (tbp) curve - pour point - flash and fire point - astm distillation curve - knocking -octane number, cetane number - smoke point) experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. summarize the manufacturing processes and applications of various should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize the treatment techniques for petroleum fractions.
            Identify the physical and chemical impurities in
  M3.01                                                         2
Understanding
            petroleum fractions.
            Summarize the purification and treatment
  M3.02                                                         6
Understanding
            processes of petroleum fractions.
            Summarize the important properties of petroleum
  M3.03                                                         2
Understanding
            products.

            Outline the experimental methods for determining
  M3.04                                                         5
Understanding
            the properties of petroleum products.

Contents:
Physical and chemical impurities in petroleum fractions.
Purification and treatment process - sweetening process of petroleum fractions -
LPG Merox
treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment
for lube oil.
Properties of petroleum fractions (API gravity - Watson Characterization factor -
Viscosity -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

support notes.

MODULE 4

petrochemicals.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: petrochemicals.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Outline the feedstock for petrochemicals.

Outline the feedstock for petrochemicals. is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the feedstock for petrochemicals. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the feedstock for petrochemicals. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the feedstock for petrochemicals. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Outline the feedstock for petrochemicals. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane,

Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane, is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the applications of petrochemicals 3 understanding illustrate the chemicals produced from methane, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding synthesis gas and olefins.

3 Understanding synthesis gas and olefins. is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding synthesis gas and olefins. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding synthesis gas and olefins. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding synthesis gas and olefins. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding synthesis gas and olefins. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/102/103102022/

Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/102/103102022/ is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/102/103102022/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend the chemicals from aromatics. 4 understanding feedstock for petrochemicals – list the major gas, liquid and solid feedstock's for petrochemical industry - identify the products from above feedstock's. list the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. production of methanol by oxidation of methane – methanol from synthesis gas – production of formaldehyde from methanol. illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - list the propylene derivatives – illustrate the production of acrylic acid and methyl methacrylate (mma). illustrate the manufacturing process for phenol, polystyrene group activity (prepare a report on the manufacturing of caprolactam in fact and compare the production method with other available manufacturing methods) text / reference t/r book title/author t1 bulk chemicals from petroleum; b.k. bhaskarao r1 modern petroleum refining processes; b. k. bhaskarao r2 petroleum refinery engineering; w.l. nelson, r3 elements of petroleum refinery engineering; o.p. gupta r4 elements of petrochemical engineering; saikat maitra & o.p. gupta online resources sl.no website link 1 https://nptel.ac.in/courses/103/102/103102022/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock’s for petrochemical industry - Identify the products from above feedstock’s. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/102/103102022/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock’s for petrochemical industry - Identify the products from above feedstock’s. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author

T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/102/103102022/ □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
petrochemicals.
  M4.01     Outline the feedstock for petrochemicals.               4
Understanding
  M4.02     Summarize the applications of petrochemicals           3
Understanding
            Illustrate the chemicals produced from methane,
  M4.03                                                            3
Understanding
            synthesis gas and olefins.
  M4.04     Comprehend the chemicals from aromatics.               4
Understanding

            Series Test – II

Contents:
Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock’s
for
petrochemical industry - Identify the products from above feedstock’s.
List the major applications of petrochemicals - polyethylene, polypropylene,
polyester,
polybutadiene, polyacrylate.
Production of methanol by oxidation of methane – Methanol from synthesis gas –
production of formaldehyde from methanol. Illustrate the manufacturing of
ethylene and
propylene by thermal cracking – manufacturing of polyethylene - List the
propylene
derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Classify fossil fuels. 2 Remembering Summarize the production methods,. (3 marks)

Answer: Begin with the standard definition or identity of *Classify fossil fuels. 2 Remembering Summarize the production methods,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding crude petroleum. Illustrate the upstream processes for crude. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding crude petroleum. Illustrate the upstream processes for crude*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the

production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil – Properties of crude oil - API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding fractional distillation. Outline various cracking and reforming processes. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding fractional distillation. Outline various cracking and reforming processes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding processes. Summarise Alkylation, Isomerisation and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding processes. Summarise Alkylation, Isomerisation and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 3

Define or explain 2 Understanding petroleum fractions. Summarize the purification and treatment. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding petroleum fractions. Summarize the purification and treatment*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding products. Outline the experimental methods for determining. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding products. Outline the experimental methods for determining*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP)

curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking - Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain Outline the feedstock for petrochemicals.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the feedstock for petrochemicals.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane,. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding synthesis gas and olefins.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding synthesis gas and olefins.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock’s for petrochemical industry - Identify the products from above feedstock’s. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/102/103102022/. (3 marks)

Answer: Begin with the standard definition or identity of *Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock’s for petrochemical industry - Identify the products from* *above feedstock’s. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/102/103102022/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Classify fossil fuels. 2 Remembering Summarize the production methods,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding petroleum fractions. Summarize the purification and treatment**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Outline the feedstock for petrochemicals.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.

- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://nptel.ac.in/courses/103/102/103102022/

IN-PAGE SEARCH

Search this lesson